

BROCKEN, Germany

WMO: 104530

Lat: 51.7986N

Lon: 10.6183E

Elev: 3781

StdP: 12.80

Time Zone: 1.00 (EUC)

Period: 82-03

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
2	2.7	7.7	-12.8	3.2	30.8	-5.1	4.8	15.8	72.6	31.7	66.9	29.8	27.0	0	1.208

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
8	9.0	68.7	58.2	66.0	56.8	63.2	55.4	60.0	65.9	58.2	63.6	56.5	61.6	17.6	210

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
57.6	81.6	62.4	55.7	76.1	60.3	54.2	72.0	58.9	28.2	66.4	26.9	63.7	25.7	61.8	68.4

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 64.3	57.5	51.8	2.4	74.8	9.3	3.2	-4.3	77.1	-9.7	79.0	-14.9	80.7	-21.7	83.0		
(5)			2.6	63.7	8.9	2.4	-3.8	65.5	-9.0	66.9	-13.9	68.2	-20.4	70.0		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	37.8	24.6	23.1	28.9	33.8	43.0	47.8	52.1	52.4	46.1	40.4	32.1	28.8		
	DBStd	12.56	8.49	9.27	7.64	7.51	7.68	7.45	7.11	6.77	6.38	7.14	7.81	6.62		
	HDD50	4831	788	754	654	486	241	131	62	53	155	310	539	658		
	HDD65	9916	1253	1174	1119	935	681	517	402	394	567	763	988	1123		
	CDD50	395	0	0	0	1	25	65	127	127	37	12	1	0		
	CDD65	6	0	0	0	0	0	1	2	3	0	0	0	0		
	CDH74	10	0	0	0	0	0	0	2	8	0	0	0	0		
	CDH80	0	0	0	0	0	0	0	0	0	0	0	0	0		
	WSAvg	25.3	35.4	29.0	26.1	21.2	19.0	19.4	19.7	20.2	24.2	28.3	29.3	32.2		
	PrecAvg	67.8	7.4	5.5	5.6	4.2	5.0	4.7	5.6	5.1	5.4	6.1	6.7	8.0		
(15) Precipitation	PrecMax	107.3	17.3	13.0	14.2	12.0	12.5	10.3	16.9	10.9	20.0	17.8	13.5	20.3		
	PrecMin	42.5	1.6	1.1	1.3	0.9	1.5	1.8	0.5	1.3	1.7	1.4	0.5	1.4		
	PrecStd	13.5	3.6	2.8	3.0	2.3	2.4	1.7	3.1	2.2	3.3	3.3	3.1	4.0		
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	42.1	47.3	52.8	55.7	63.6	70.4	72.0	74.2	64.7	61.5	53.5	47.0		
		MCWB	35.7	36.6	40.8	45.7	52.0	56.8	60.5	60.4	55.4	49.8	44.3	36.9		
	2%	DB	39.2	42.6	47.4	51.5	60.3	65.9	68.4	68.7	61.9	57.5	49.7	43.7		
		MCWB	35.8	36.5	38.7	44.1	50.4	56.7	58.9	57.9	54.1	49.3	41.9	35.2		
	5%	DB	37.0	39.0	42.6	48.1	57.5	62.4	66.1	65.3	59.2	53.8	46.4	41.0		
		MCWB	34.8	34.6	38.5	42.1	49.2	54.9	57.2	56.6	53.0	47.9	40.3	36.1		
	10%	DB	34.0	35.6	39.1	45.1	54.7	59.1	63.2	62.3	55.8	50.5	43.3	38.6		
		MCWB	31.9	33.0	37.0	40.5	47.7	53.4	55.8	55.3	51.7	46.4	40.0	35.5		
	0.4%	WB	39.6	42.0	43.1	47.2	54.0	61.0	63.0	62.6	58.1	53.9	46.1	41.9		
		MCDB	39.9	43.6	48.5	52.7	59.8	66.2	69.1	71.9	61.9	57.2	48.7	43.2		
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	2%	WB	37.5	38.5	40.8	45.2	52.0	57.7	60.4	59.4	55.8	51.9	44.5	40.1		
		MCDB	37.8	40.3	44.2	49.6	58.0	63.2	66.1	65.8	59.4	56.0	46.9	41.1		
	5%	WB	35.0	35.0	39.1	43.3	50.1	55.6	58.5	57.7	54.1	49.3	42.6	37.8		
		MCDB	36.1	37.4	42.1	47.4	56.1	61.4	64.0	63.3	57.2	52.0	45.1	39.3		
	10%	WB	32.4	33.1	36.9	40.9	48.4	53.7	56.7	56.2	52.0	47.3	40.6	35.7		
		MCDB	33.1	34.9	39.2	44.6	53.6	58.4	62.0	61.1	54.8	49.3	42.7	37.3		
(35) Mean Daily Temperature Range	5% DB	MDBR	7.0	7.2	6.9	8.2	9.1	8.8	9.5	9.0	7.6	7.4	6.7	6.7		
		MCDBR	9.9	9.8	10.1	11.2	11.5	11.9	12.1	12.7	11.9	10.4	10.2	10.5		
		MCWBR	8.9	7.6	7.3	7.4	7.0	8.3	8.4	7.3	8.2	7.3	7.6	8.1		
		MCDBR	8.8	9.0	9.2	10.7	11.4	11.2	11.9	11.5	10.4	9.2	8.6	8.5		
	5% WB	MCWBR	8.7	8.1	7.4	8.0	7.9	8.3	9.1	7.5	8.1	7.5	8.1	8.1		
(40) Clear-Sky Solar Irradiance	taub		0.268	0.286	0.316	0.344	0.347	0.345	0.358	0.352	0.325	0.304	0.287	0.264		
	taud		2.468	2.459	2.410	2.369	2.373	2.386	2.346	2.383	2.465	2.522	2.509	2.459		
	Ebn at Noon		245	271	281	284	287	288	281	278	274	260	234	226		
	Edh at Noon		17	23	29	35	36	36	37	34	28	22	16	15		
(44) All-Sky Solar Radiation	RadAvg		223	443	786	1269	1556	1680	1577	1345	978	562	258	170		
	RadStd		26	61	89	153	180	138	212	123	121	61	32	18		

Historical Trends

		DBAvg		Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65		
Station Only		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	
		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	
Regional (44 neighbors)		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	

Nomenclature: See separate page